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Q1 Define Statistics

Definition

Statistics is defined as the science of **collecting, organising, presenting, analysing, and interpreting numerical data** to assist in making sound and rational decisions. The word 'Statistics' is derived from the Latin word 'Status' meaning the state. It was originally used by governments to collect data on population and trade.

Definitions by Scholars

Scholar	Definition
Croxton and Cowden	Statistics is the science which deals with collection, analysis and interpretation of numerical data.
A.L. Bowley	Statistics may be called the science of counting or the science of averages.
Horace Secrist	Statistics is the aggregate of facts affected to a marked extent by a multiplicity of causes.

Two Meanings of Statistics

Singular Sense (Statistical Methods): Refers to the subject or discipline itself, i.e., the methods used to deal with numerical data. Example: The study of statistics as a subject.

Plural Sense (Statistical Data): Refers to numerical facts or data collected on any topic. Example: Population statistics, crime statistics, production statistics.

Scope and Branches of Statistics

Branch	Focus	Examples
Descriptive Statistics	Summarises and describes raw data using tables, charts, averages.	Mean, Median, Mode, Graphs
Inferential Statistics	Draws conclusions about a population based on a sample.	Hypothesis tests, Confidence Intervals
Applied Statistics	Applies statistical methods to real-world problems.	Business forecasting, Quality Control

Functions of Statistics

- (i) Simplification of Complex Data: Reduces large volumes of data to simple averages and summaries.
- (ii) Facilitates Comparison: Enables comparison across time periods, regions, or products.
- (iii) Helps in Forecasting: Trends and patterns derived from past data help predict future values.
- (iv) Tests Hypotheses: Statistical tests verify claims and assumptions about populations.

- (v) Formulates Policies: Governments use census and survey data for planning and policy-making.
- (vi) Measures Relationships: Correlation and regression tools measure how variables relate to each other.

Limitations of Statistics

- (i) Statistics deals only with quantitative (numerical) data; qualitative facts are not directly measurable.
- (ii) Statistical results are true only on average, not for individual cases.
- (iii) Statistical methods can be misused to draw misleading conclusions.
- (iv) Statistics does not study the causes of phenomena, only their numerical patterns.

Q2 What is Regression Analysis?

Definition

Regression Analysis is a **statistical technique used to examine the relationship between one dependent variable and one or more independent variables**, and to predict the value of the dependent variable based on the independent variable(s). The term 'Regression' was first used by Sir Francis Galton in 1877 while studying the relationship between heights of parents and children.

Simple Linear Regression Equation

$$\hat{Y} = a + bX$$

$\hat{Y}$ = Predicted value of the dependent variable

X = Value of the independent variable

a = Y-intercept: value of Y when X = 0

b = Regression coefficient (slope): change in Y for a unit change in X

Formulas for Regression Coefficients

Regression of Y on X: $b_{yx} = r \times (S_y / S_x)$ and $a = \bar{Y} - b \times \bar{X}$

Regression of X on Y: $b_{xy} = r \times (S_x / S_y)$ and $a = \bar{X} - b \times \bar{Y}$

where r = Karl Pearson correlation coefficient, S_y = SD of Y, S_x = SD of X

Types of Regression

Type	Description	Example
Simple Linear Regression	One independent variable, straight-line relationship	Sales vs. Advertising spend
Multiple Linear Regression	Two or more independent variables	Sales vs. Price, Income, Advertising

Type	Description	Example
Logistic Regression	Dependent variable is binary (0 or 1)	Pass/Fail prediction based on study hours
Polynomial Regression	Curved relationship between variables	Population growth over time
Non-linear Regression	Relationship follows a non-linear equation	Diminishing returns in production

Properties of Regression Lines

- (i) The two regression lines always intersect at the point ($\bar{X}$, $\bar{Y}$).
- (ii) Regression line of Y on X is used to predict Y for a given value of X.
- (iii) Regression line of X on Y is used to predict X for a given value of Y.
- (iv) If $r = 1$ or -1 (perfect correlation), both regression lines coincide.
- (v) If $r = 0$, regression lines are perpendicular to each other.
- (vi) The product of regression coefficients = r^2 : $b_{yx} \times b_{xy} = r^2$

Numerical Example

Given: $\bar{X} = 6$, $\bar{Y} = 9$, SD of X = 2, SD of Y = 3, $r = 0.8$

$$b_{yx} = r \times (S_y/S_x) = 0.8 \times (3/2) = 1.2$$

$$a = \bar{Y} - b \times \bar{X} = 9 - 1.2 \times 6 = 9 - 7.2 = 1.8$$

$$\text{Regression Equation: } \hat{Y} = 1.8 + 1.2X$$

Interpretation: For every unit increase in X, Y increases by 1.2 units.

Difference Between Correlation and Regression

Concept	Purpose	Usage
Correlation	Measures strength and direction of relationship	Does not predict values
Regression	Establishes mathematical equation	Used to predict unknown values

Q3 State the Properties of Arithmetic Mean

Definition

The **Arithmetic Mean (AM)** is the most commonly used measure of central tendency. It is calculated by dividing the sum of all observations by the total number of observations. It is also called the average or simply the mean.

$$\text{Formula: } \bar{X} = \text{Sum of all values (SX)} / \text{Number of values (N)}$$

Properties of Arithmetic Mean

Property	Explanation	Example
1. Uniqueness	There is only one arithmetic mean for a given set of data. It is uniquely determined.	Data: 2,4,6 -> Mean = 4 (only one value)
2. Uses Every Value	The mean takes into account every single value in the dataset. No observation is ignored.	Data: 1,2,3,4,5 -> Mean = 3 (all 5 used)
3. Sum of Deviations = 0	The algebraic sum of deviations of all values from the mean is always zero: $\sum (X_i - \bar{X}) = 0$	Values: 2,3,4 Mean=3; $(2-3)+(3-3)+(4-3) = -1+0+1 = 0$
4. Minimum Sum of Squares	The sum of squared deviations from the mean is less than from any other value (Least Squares Property).	$\sum (X_i - \bar{X})^2$ is minimum
5. Combined Mean	If means of two groups are known, a combined mean can be found: $\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$	$n_1=10, \bar{X}_1=20; n_2=20, \bar{X}_2=25$ -> $\bar{X} = (200+500)/30 = 23.33$
6. Change of Origin	If $Y = X + A$ (shifting), then $\bar{Y} = \bar{X} + A$	If each value increases by 5, mean also increases by 5
7. Change of Scale	If $Y = kX$ (scaling), then $\bar{Y} = k * \bar{X}$	If each value doubles, the mean also doubles
8. Affected by Extremes	Extreme values (outliers) pull the mean towards them, making it an unreliable measure for skewed data.	Data: 1,2,3,4,100 -> Mean=22 (misleading due to 100)

Merits and Demerits of Arithmetic Mean

Type	Points
MERITS	Easy to calculate and understand; Uses all data; Suitable for further algebra; Stable measure.
DEMERITS	Affected by extreme values; Cannot be calculated if any value is unknown; Not suitable for open-end classes.

Q4 Define Binomial Distribution

Definition

A **Binomial Distribution** is a **discrete probability distribution** that describes the number of successes in a fixed number of independent trials, where each trial has only two outcomes: Success or Failure. It was first derived by James Bernoulli in 1700.

Conditions: (i) Fixed number of trials (n), (ii) Each trial is independent, (iii) Only two outcomes per trial, (iv) Probability of success (p) is constant in every trial.

Probability Mass Function (PMF)

$$P(X = r) = nCr \times p^r \times q^{(n-r)}$$

n = Total number of independent trials

r = Number of successes required (r = 0, 1, 2, ..., n)

p = Probability of success in a single trial

q = 1 - p = Probability of failure in a single trial

$nCr = n! / [r! \times (n-r)!]$ (number of combinations)

Parameters and Statistical Measures

Measure	Formula	Numerical Example
Mean (Expected Value)	$E(X) = np$	n=10, p=0.3 -> Mean = 3
Variance	$Var(X) = npq$	n=10, p=0.3, q=0.7 -> Var = 2.1
Standard Deviation	$SD = \sqrt{npq}$	$SD = \sqrt{2.1} = 1.449$
Skewness	$(q - p) / \sqrt{npq}$	Symmetric when p = 0.5
Kurtosis	$3 + (1-6pq)/(npq)$	Approaches Normal as n increases

Properties of Binomial Distribution

- (i) The distribution is symmetric when p = 0.5; positively skewed when p < 0.5; negatively skewed when p > 0.5.
- (ii) Sum of all probabilities = 1: Sum of P(X=r) for r=0 to n = 1.
- (iii) As n increases and p is small, Binomial approaches Poisson Distribution.
- (iv) As n becomes large, Binomial approaches Normal Distribution.
- (v) The mode is the most probable value of r.

Solved Example 1

A fair coin is tossed 5 times. Find the probability of getting exactly 3 heads.

Given: n = 5, r = 3, p = 0.5 (Head), q = 0.5 (Tail)

$$P(X=3) = 5C3 \times (0.5)^3 \times (0.5)^2 = 10 \times 0.125 \times 0.25 = 0.3125$$

Solved Example 2

A batch of bulbs has 20% defective items. If 4 bulbs are selected randomly, find the probability that exactly 2 are defective.

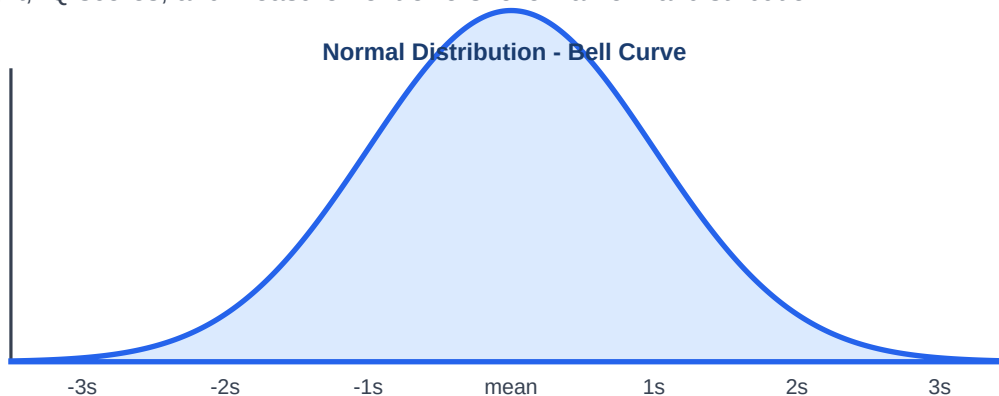
Given: n = 4, r = 2, p = 0.20, q = 0.80

$$P(X=2) = 4C2 \times (0.2)^2 \times (0.8)^2 = 6 \times 0.04 \times 0.64 = 0.1536$$

Q5 Define Normal Distribution

Definition

The **Normal Distribution** is a **continuous probability distribution** that is perfectly symmetric about its mean and follows a bell-shaped curve. It is the most important distribution in statistics, introduced by Carl Friedrich Gauss (also called Gaussian distribution). Most natural phenomena such as height, weight, IQ scores, and measurement errors follow a normal distribution.



Probability Density Function

$$f(x) = \frac{1}{(s \times \sqrt{2\pi})} \times e^{-\frac{(x-m)^2}{(2s^2)}}$$

m (μ) = Mean of the distribution

s (σ) = Standard deviation

p (π) = 3.14159

e = Euler's number = 2.71828

x = Any real value of the random variable

Properties of Normal Distribution

Property	Explanation
Bell-shaped Curve	The graph is symmetric and bell-shaped, peaking at the mean.
Mean = Median = Mode	All three measures of central tendency coincide at the centre.
Symmetry	The curve is perfectly symmetric about the mean. Left half = Right half.
Empirical Rule (68-95-99.7)	68% values lie within 1 SD; 95% within 2 SD; 99.7% within 3 SD of mean.

Property	Explanation
Total Area = 1	The total area under the normal curve is exactly 1 (represents 100% probability).
Asymptotic Tails	The tails of the curve extend to infinity but never touch the x-axis.
Unimodal	The distribution has only one peak (one mode), at the mean.
Defined by m and s	The distribution is completely described by just two parameters: mean and SD.

Standard Normal Distribution

When a normal variable X is converted to Z using the formula below, it follows the Standard Normal Distribution with mean = 0 and SD = 1.

$$Z = (X - m) / s$$

Z-tables (Standard Normal Tables) are used to find probabilities.

Applications

- (i) Quality Control: Products with measurements within 2 SD of mean are accepted.
- (ii) Finance: Returns on stocks are approximately normally distributed.
- (iii) Education: Exam scores in large populations follow a normal distribution.
- (iv) Medicine: Drug dosage effects are normally distributed across patients.

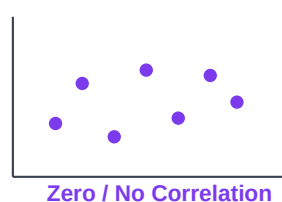
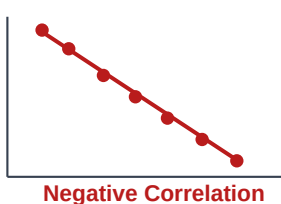
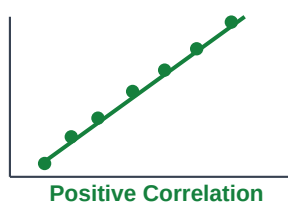
Q6 What are the Different Types of Correlation?

Definition

Correlation is a statistical measure that quantifies the **strength and direction of the linear relationship between two variables**. When a change in one variable is accompanied by a corresponding change in another, the two variables are said to be correlated. Karl Pearson developed the concept of the correlation coefficient (r).

The correlation coefficient r always lies between -1 and +1: $-1 \leq r \leq +1$

Types of Correlation - Scatter Diagrams



Classification A: Based on Direction

Type	Meaning	Range of r	Example
Positive Correlation	Both variables move in the SAME direction. When X increases, Y also increases.	$r = 0 \text{ to } +1$	Income and Expenditure; Temperature and Ice cream sales; Study hours and Marks
Negative Correlation	Variables move in OPPOSITE directions. When X increases, Y decreases.	$r = -1 \text{ to } 0$	Price and Demand; Speed and Travel time; Exercise and Body weight
Zero / No Correlation	No linear relationship exists between the two variables.	$r = 0$	Shoe size and Intelligence; Hair colour and Salary

Classification B: Based on Number of Variables

Type	Meaning	Example
Simple Correlation	Relationship between exactly two variables only.	Weight and Height
Multiple Correlation	Relationship of one variable with two or more others simultaneously.	Sales depending on Price, Advertising and Income
Partial Correlation	Relationship between two variables while keeping all others constant.	Y vs. X1 while holding X2 constant

Classification C: Based on Linearity

Type	Meaning	Example
Linear Correlation	When plotted, points fall approximately on a straight line. The ratio of change is constant.	Proportional increase in sales with advertising budget
Non-Linear (Curvilinear)	Points fall on a curve, not a straight line. Rate of change is not constant.	Diminishing returns: output vs. workers added

Degrees of Correlation

Degree	Value of r	Interpretation
Perfect Positive	$r = +1$	Both variables move exactly together
High Positive	$r = 0.75 \text{ to } 1$	Strong positive relationship
Moderate Positive	$r = 0.50 \text{ to } 0.75$	Moderate positive relationship
Low Positive	$r = 0 \text{ to } 0.50$	Weak positive relationship

Degree	Value of r	Interpretation
No Correlation	$r = 0$	No linear relationship
Low Negative	$r = 0$ to -0.50	Weak negative relationship
High Negative	$r = -0.75$ to -1	Strong negative relationship
Perfect Negative	$r = -1$	Both variables move in exactly opposite directions

Q7 Objectives of Time Series Analysis and Variations

What is Time Series?

A **Time Series** is a sequence of observations or data points recorded at successive points in time, usually at equal intervals (daily, monthly, quarterly, yearly). The data is arranged in chronological order. Examples include annual rainfall, monthly sales figures, daily stock prices, etc.

Objectives of Studying Time Series Analysis

Objective	Explanation	Example
1. Understanding Past Behaviour	Helps identify patterns and trends in historical data to understand how a variable has changed over time.	Analysing sales data over 10 years to understand growth phases
2. Forecasting Future Values	Extrapolates historical patterns to predict future values of the variable accurately.	Predicting next year's GDP based on past 20 years of data
3. Planning and Decision Making	Provides a scientific basis for managerial planning, budgeting, and resource allocation.	A company plans production levels based on demand forecast
4. Comparison Across Time	Facilitates comparison of data across different time periods to measure progress.	Comparing quarterly sales of 2023 vs. 2024
5. Evaluation of Policies	Measures the effectiveness of government policies or business strategies over time.	Effect of demonetisation on GDP growth
6. Isolation of Components	Separates the effects of trend, seasonal, cyclical and random components for better analysis.	Removing seasonality from monthly sales to see true trend

Variations (Fluctuations) in Time Series

Component	Description	Example
Secular Trend (T)	A long-term, smooth upward or downward movement observed over a long period of time (decades). Reflects the general direction of growth or decline.	India's GDP has been rising steadily over 30 years; Population growth
Seasonal Variations (S)	Regular and predictable fluctuations that occur within a year due to seasons, festivals, customs, or weather. They repeat every year at the same time.	Umbrella sales rise during monsoon; Sweets sales peak during Diwali
Cyclical Variations (C)	Medium to long-term oscillations lasting 2 to 10 years, associated with the business cycle phases: Prosperity (Boom), Recession, Depression, and Recovery.	Stock market boom of 2006-07 followed by recession of 2008-09
Irregular / Random Variations (I)	Completely unpredictable and erratic short-term movements caused by chance or unforeseen events like wars, earthquakes, floods, or pandemics. Cannot be modelled or forecast.	Drop in airline bookings due to COVID-19 in March 2020

Q8 Define Smoothing Methods and Name Them

Definition

Smoothing Methods are **statistical techniques applied to time series data to remove random fluctuations (noise) and highlight the underlying trend or pattern**. They work by averaging data values in various ways so that the irregular component is reduced, making it easier to identify the true direction of the data. These methods are widely used in business forecasting, demand planning, and financial analysis.

Types / Names of Smoothing Methods

Method	Description	Example	Best Used When
1. Simple Moving Average (SMA)	Computes the arithmetic mean of a fixed number (window) of consecutive observations. The window moves forward one period at a time.	3-year MA: Average of years 1,2,3; then 2,3,4; then 3,4,5...	Easy to calculate; removes short-term fluctuations
2. Weighted Moving Average (WMA)	Similar to SMA but assigns different weights to different periods. More recent observations are given higher weights.	Weights: 0.5 (recent), 0.3 (prev), 0.2 (oldest)	Gives more importance to recent data

Method	Description	Example	Best Used When
3. Simple Exponential Smoothing	Uses smoothing constant α (alpha). Each forecast is a weighted average of the actual value and previous forecast: $S_t = \alpha X_t + (1-\alpha)S_{t-1}$	$\alpha = 0.3$; if last actual=100 and last forecast=95: $S_t = 0.3*100 + 0.7*95 = 96.5$	Suitable for data with NO trend or seasonality
4. Double Exponential Smoothing (Holt's Method)	Applies exponential smoothing TWICE to handle data with a LINEAR TREND. Uses two smoothing constants: α (level) and β (trend).	Used when sales data shows consistent increase each year	Best for trended data without seasonality
5. Triple Exponential Smoothing (Holt-Winters Method)	Extends Holt's method to handle both TREND and SEASONALITY using three smoothing constants.	Monthly retail sales with seasonal peaks at year-end	Best for trended AND seasonal data
6. Least Squares Trend Line	Fits a mathematical straight line to the data: $Y = a + bX$, where a and b are calculated to minimize sum of squared errors.	Annual production data fitted with $Y = 200 + 15X$	Most accurate method; gives equation for future prediction

Key Rule: Smoothing Constant (α): $0 < \alpha < 1$

High α (close to 1): More weight to recent data - forecast reacts quickly to changes.

Low α (close to 0): More weight to older data - forecast is stable and smooth.

Moving Average Example (3-Year MA)

Year	Actual Sales	Calculation	3-Year Moving Average
2019	100	-	-
2020	110	$(100+110+120)/3$	110.0
2021	120	$(110+120+130)/3$	120.0
2022	130	$(120+130+150)/3$	133.3
2023	150	-	-

Q9 Define Hypothesis with an Example

Definition

A **Hypothesis** is a **tentative, testable statement or assumption made about a population parameter or about the relationship between variables**, which can be verified or disproved through statistical testing. It is a preliminary explanation or conjecture based on limited evidence that serves as a starting point for an investigation.

Types of Hypothesis

Type	Explanation	Example
Null Hypothesis (H ₀)	States that there is NO significant difference, relationship, or effect. It is the default assumption that we seek to disprove.	H ₀ : The new drug has no effect on blood pressure. ($m = 120$ mmHg)
Alternative Hypothesis (H ₁ or H _a)	States that there IS a significant difference, relationship, or effect. It is what the researcher seeks to prove.	H ₁ : The new drug reduces blood pressure. ($m < 120$ mmHg)
Simple Hypothesis	Completely specifies the value of the population parameter.	H ₀ : $p = 0.5$ (probability of heads is exactly 0.5)
Composite Hypothesis	Does not completely specify the parameter; gives a range or direction.	H ₁ : $m > 50$ or H ₁ : $m \neq 50$
Directional (One-tailed)	Specifies the direction of the difference (greater than or less than).	H ₁ : $m > 100$ or H ₁ : $m < 100$
Non-Directional (Two-tailed)	Does not specify the direction; only says the parameter is different.	H ₁ : $m \neq 100$

Detailed Practical Example 1 - Business

Situation: A company claims its new training programme improves average productivity above 80 units/day.

H₀: $m = 80$ (Training has NO effect; productivity is 80 units/day)

H₁: $m > 80$ (Training INCREASES productivity above 80 units/day)

Test: One-tailed t-test or Z-test at $\alpha = 0.05$

Decision: If sample mean $>$ critical value, Reject H₀ and accept H₁.

Conclusion: There is sufficient evidence that training improves productivity.

Practical Example 2 - Quality Control

Situation: A machine fills bottles with mean 500ml. Inspector suspects it is over or under-filling.

H₀: $m = 500$ ml (Machine is working correctly)

H1: $\mu \neq 500$ ml (Machine is NOT filling correctly - two-tailed)

Test: Two-tailed Z-test at $\alpha = 0.05$

Decision: If $|Z\text{-calculated}| > Z\text{-critical}$ (1.96), Reject H_0 .

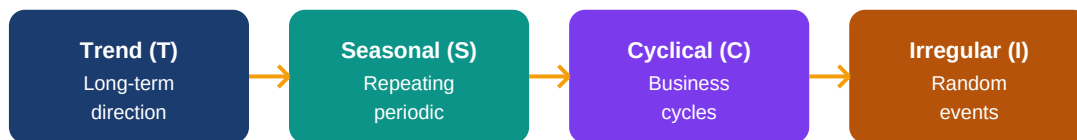
Conclusion: Machine needs recalibration.

Q1 0 Explain Components of Time Series

Definition of Time Series

A **Time Series** is an ordered sequence of values of a variable recorded at successive points in time. The main purpose of decomposing a time series is to understand the behaviour of data over time and to improve forecasting accuracy.

Components of Time Series: $Y = T \times S \times C \times I$



1. Secular Trend (T)

Secular Trend refers to the **long-term, smooth upward or downward movement in data** over a long period of time (several years or decades). It reflects the general direction of growth or decline and is caused by fundamental economic, social, or technological factors.

Example 1: India's population has shown a consistent upward trend over the past 70 years.

Example 2: The cost of electronic devices (e.g., mobile phones) has shown a downward trend due to technology.

Example 3: Life expectancy in developed countries has shown an increasing trend over 50 years.

2. Seasonal Variations (S)

Seasonal Variations are **regular, periodic fluctuations that recur within a period of one year**. They are caused by seasonal factors such as climate, customs, festivals, or social habits. These variations are predictable and repeat every year at approximately the same time.

Example 1: Sales of raincoats and umbrellas peak during the monsoon season (June-September).

Example 2: Demand for woollen clothes increases during winter (November-January).

Example 3: Sales of sweets and gifts increase during Diwali and Christmas festivals.

3. Cyclical Variations (C)

Cyclical Variations are **medium to long-term oscillations in data that are related to the business cycle**. They typically last 2 to 10 years and pass through four phases: Prosperity (Boom), Recession,

Depression, and Recovery. Unlike seasonal variations, they are not fixed to a calendar year.

Example 1: Economic boom of 2005-2007 followed by the global recession of 2008-2009.

Example 2: The IT industry experienced growth in 1990s, bust in 2000 (dot-com crash), then recovery.

Example 3: Real estate prices follow cyclical patterns with boom and bust phases.

4. Irregular / Random Variations (I)

Irregular or Random Variations are **completely unpredictable and erratic short-term fluctuations** in data that cannot be explained by trend, seasonal, or cyclical patterns. They are caused by chance events, accidents, natural disasters, or sudden policy changes. These cannot be forecasted or modelled.

Example 1: Sudden crash in stock markets due to the COVID-19 pandemic in March 2020.

Example 2: Drop in production in a factory due to an unexpected machine breakdown.

Example 3: Spike in demand for essential commodities due to a flood warning.

Models of Time Series

Model	Formula	When Used
Multiplicative Model	$Y = T \times S \times C \times I$	Used when components interact proportionally; most common in business
Additive Model	$Y = T + S + C + I$	Used when components act independently and have constant magnitudes

Q1 1 Define Time Series Analysis and Mention the Methods

Definition

Time Series Analysis is the process of **examining a set of data points collected over time to identify meaningful patterns, trends, seasonal effects, and cycles**. The goal is to understand the underlying structure of the data so that accurate forecasts can be made for future time periods. It is extensively used in economics, business, meteorology, finance, and engineering.

Importance of Time Series Analysis

- (i) Enables accurate forecasting of future demand, production, prices, and other variables.
- (ii) Helps businesses plan inventories, budgets, and workforce requirements.
- (iii) Aids economists and governments in tracking economic indicators like GDP and inflation.
- (iv) Helps identify seasonal effects and plan marketing and promotional activities accordingly.

Methods of Measuring Trend in Time Series

Method	Description	Merits / Demerits	Best Used For
1. Graphic / Free-Hand Curve Method	Data is plotted on a graph and a smooth trend curve is drawn by inspection and judgment.	Advantages: Simple and quick. Disadvantages: Subjective; varies from person to person.	Quick preliminary analysis
2. Semi-Average Method	The data is divided into two equal halves. The mean of each half is computed and plotted. A straight line connecting the two points is the trend line.	Advantages: Objective, simple. Disadvantages: Affected by extreme values.	Small datasets with linear trend
3. Moving Average Method	Calculates successive averages over a rolling window of periods (3-year, 5-year). Smooths out short-term fluctuations.	Advantages: Simple, removes fluctuations. Disadvantages: Loses data at ends.	Identifying long-term trend
4. Least Squares Method	Fits a mathematical equation $Y = a + bX$ that minimises the sum of squared deviations between actual and predicted values. Values of a and b are computed using normal equations.	Advantages: Most accurate, provides forecast equation. Disadvantages: Assumes linearity.	Precise forecasting and trend measurement
5. Exponential Smoothing	Uses a smoothing constant (a) to give more weight to recent observations and less to older ones. Formula: $St = a \cdot Xt + (1-a) \cdot St-1$	Advantages: Reacts quickly to changes; good for short-term forecasts.	Short-term business forecasting

Normal Equations for Least Squares Method

$$\text{Sum of } Y = N \cdot a + b \cdot \text{Sum}(X)$$

$$\text{Sum of } XY = a \cdot \text{Sum}(X) + b \cdot \text{Sum}(X^2)$$

When origin is at the middle of time series: $\text{Sum}(X) = 0$, so $a = \text{Sum}(Y)/N$ and $b = \text{Sum}(XY)/\text{Sum}(X^2)$

Q1
2

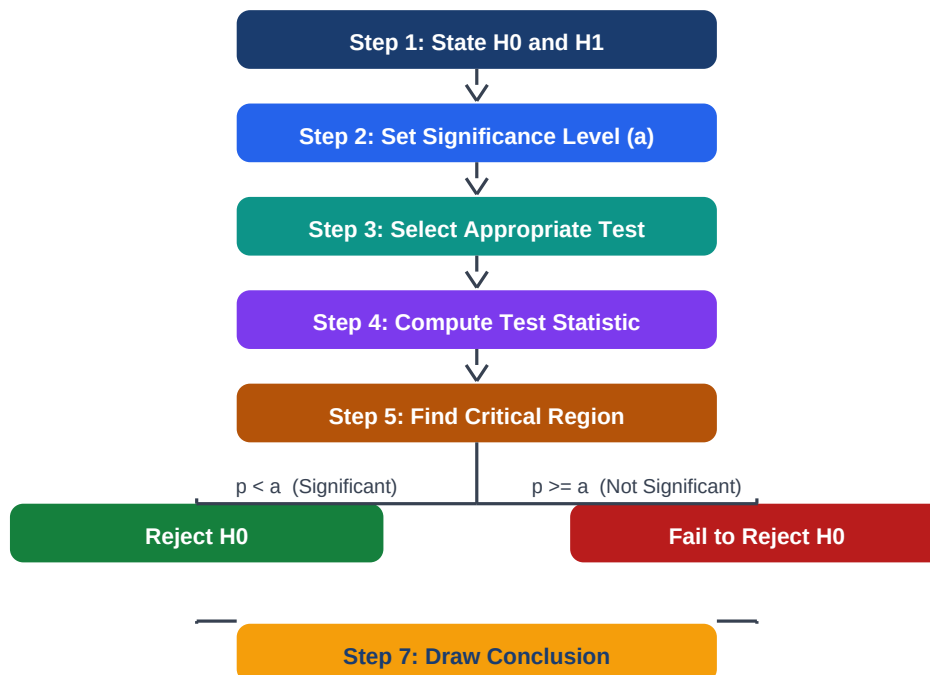
Explain the Flow of Hypothesis Testing with Diagram

Definition

Hypothesis testing is a **formal statistical procedure used to make decisions about population parameters based on sample data**. It helps determine whether the evidence from the sample is strong enough to reject a stated assumption (null hypothesis) about the population.

Detailed Steps in Hypothesis Testing

Step	Name	Action	Example
Step 1	State the Hypotheses	Clearly formulate H0 (null hypothesis) and H1 (alternative hypothesis).	H0: $\mu = 50$ marks; H1: $\mu \neq 50$ marks
Step 2	Set Level of Significance (α)	Choose the probability of making a Type I error. Common values: $\alpha = 0.05$ (5%) or $\alpha = 0.01$ (1%).	$\alpha = 0.05$ means we accept 5% risk of wrongly rejecting H0
Step 3	Choose Appropriate Test	Select the statistical test based on data type, sample size, and whether population SD is known.	Z-test ($n > 30$ or σ known), t-test ($n < 30$), Chi-square, F-test
Step 4	Collect Data and Compute Test Statistic	Gather sample data and calculate the test statistic using the relevant formula.	$Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$
Step 5	Determine Critical Region	Find the critical value from statistical tables based on α and the nature of the test.	For $\alpha = 0.05$, two-tailed: Z-critical = ± 1.96
Step 6	Make a Decision	Compare computed test statistic with critical value OR compare p-value with α .	If $ Z > 1.96$: Reject H0. If $ Z \leq 1.96$: Fail to Reject H0.
Step 7	Draw Conclusion	Interpret the statistical decision in the context of the original research problem.	Conclude whether evidence supports or rejects the claim.



Flow Chart: Steps in Hypothesis Testing

Types of Errors in Hypothesis Testing

Error Type	Definition	Also Called	Real-World Analogy
Type I Error (a)	Rejecting H ₀ when it is actually TRUE.	False Positive	Convicting an innocent person
Type II Error (b)	Failing to reject H ₀ when it is actually FALSE.	False Negative	Acquitting a guilty person

Common Statistical Tests and When to Use Them

Test	When to Use
Z-test	Test about population mean; sample size ≥ 30 or population SD is known
t-test	Test about population mean; sample size < 30 and population SD is unknown
Chi-square	Test for independence of attributes or goodness of fit (categorical data)
F-test / ANOVA	Comparing variances or means of three or more groups simultaneously
Mann-Whitney U	Non-parametric alternative to t-test for small or non-normal data

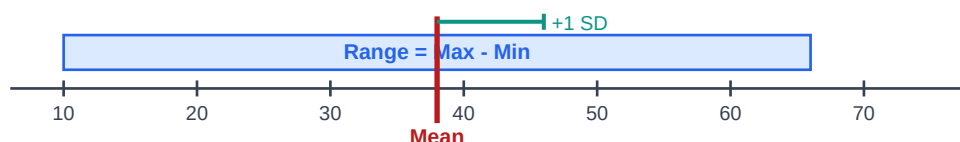
Q1
3

Explain the Significance of Measuring Dispersion

Definition of Dispersion

Dispersion (also called Variation or Spread) refers to the **extent to which the values in a dataset are scattered or spread around the central value**. While measures of central tendency (mean, median, mode) tell us where the data is centred, measures of dispersion tell us how spread out the data is. Two datasets can have the same mean but very different dispersions.

Measures of Dispersion Illustration



Significance of Measuring Dispersion

Significance	Explanation	Example
1. Reliability of Central Tendency	Low dispersion indicates the mean is a good representative of the data; high dispersion means the mean is unreliable.	Students with consistent marks (SD=3) vs. irregular marks (SD=25)
2. Comparative Study	Allows comparison of variability between two or more groups, even when means are equal.	Investment A: mean=10% SD=2% is less risky than Investment B: mean=10% SD=15%
3. Quality Control	In manufacturing, dispersion helps monitor whether products meet specifications. Low SD = uniform quality.	Machine producing bolts: if SD is too high, bolts are rejected
4. Basis for Further Statistical Analysis	SD and variance are foundation for correlation, regression, hypothesis testing, and confidence intervals.	Z-test, t-test, F-test all use SD or variance
5. Risk Measurement in Finance	Standard deviation of returns measures investment risk. Higher SD = higher financial risk.	Mutual fund with SD=5% is less volatile than one with SD=20%
6. Identifies Uniformity	Low dispersion indicates that data values are similar to each other (uniformity). High dispersion shows diversity.	Uniform wages across employees (low SD) vs. wide pay gaps (high SD)
7. Helps in Sampling	When dispersion is high, a larger sample is needed to get accurate estimates of the population.	Highly variable income data requires larger sample for surveys

Measures of Dispersion with Formulas

Measure	Formula	Remark
Range	$L - S$ (Largest - Smallest value)	Simple but affected by extremes
Mean Deviation	$MD = \frac{\sum X_i - \bar{X} }{N}$	Ignores direction of deviation
Variance	$s^2 = \frac{\sum (X_i - \bar{X})^2}{N}$	Squared units; widely used
Standard Deviation (SD)	$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{N}}$	Most popular; same units as data
Coefficient of Variation	$CV = (s / \bar{X}) \times 100$	Relative measure; unitless; for comparison
Quartile Deviation	$QD = (Q_3 - Q_1) / 2$	Used with skewed distributions

Q1
4

Discuss Difference Between Parametric and Non-Parametric Tests

Parametric Tests - Definition

Parametric tests are statistical tests that make specific assumptions about the distribution of the population from which the sample is drawn. They assume that the data follows a normal distribution and are based on population parameters such as mean (μ) and standard deviation (σ). They are generally more powerful and accurate.

Non-Parametric Tests - Definition

Non-parametric tests (also called distribution-free tests) do not require any assumptions about the underlying population distribution. They can be used with nominal, ordinal, or non-normally distributed data. They are less powerful than parametric tests but more flexible and applicable in a wider variety of situations.

Detailed Comparison Table

Basis of Comparison	Parametric Tests	Non-Parametric Tests
Assumption about Distribution	Assumes normal distribution of population	No assumption about distribution (distribution-free)
Type of Data	Interval or Ratio scale (continuous numerical data)	Nominal, Ordinal, or non-normal continuous data
Population Parameters	Based on parameters: mean, variance, SD	Does not use population parameters
Sample Size	Requires large sample size (generally $n \geq 30$)	Can work with small sample sizes ($n < 30$)
Statistical Power	Higher power - less likely to miss a real effect	Lower power than parametric tests
Computational Complexity	Mathematically complex formulas required	Simpler calculations; uses ranks or signs
Precision	Provides more precise and exact results	Less precise; results in terms of ranks or categories
Outlier Sensitivity	Highly sensitive to outliers in the data	Less sensitive to outliers
Examples of Tests	Z-test, t-test, F-test, Pearson Correlation, ANOVA	Chi-square, Mann-Whitney U, Wilcoxon, Kruskal-Wallis, Spearman rho
When to Use	When data is normally distributed and sample is large	When normality is violated or data is ordinal/nominal
Usage in Business	Testing quality control, salary comparisons between two departments	Customer satisfaction surveys, product ranking comparisons

Examples with Context

Situation	Appropriate Test
Comparing means of 2 groups, large sample, normal data	Independent samples Z-test or t-test
Comparing 3 or more group means (normal data)	ANOVA (F-test)
Testing relationship between two quantitative variables	Pearson Correlation
Comparing two groups, small sample, non-normal data	Mann-Whitney U test
Testing if observed frequencies match expected	Chi-Square Goodness of Fit
Comparing rankings of two judges	Spearman Rank Correlation

Q1 5 Define Mode and Give Two Suitable Examples

Definition

The **Mode** is defined as the value that **occurs most frequently** in a given dataset. It is the most popular or most common value in the distribution. The mode is the only measure of central tendency that can be used for nominal (categorical) data. The term 'Mode' comes from the French word 'La Mode' meaning 'fashionable'.

Formula for Grouped Data (Czuber's Interpolation Formula)

$$Mo = L + [(f1 - f0) / (2f1 - f0 - f2)] \times h$$

L = Lower boundary of the modal class (class with highest frequency)

f1 = Frequency of the modal class

f0 = Frequency of the class PRECEDING the modal class

f2 = Frequency of the class FOLLOWING the modal class

h = Class width (class interval)

Types of Mode

Type	Meaning	Example
Unimodal	Exactly one mode (one most frequent value)	2, 3, 4, 4, 5, 6 -> Mode = 4
Bimodal	Two modes (two values each with same high frequency)	1, 2, 2, 3, 3, 4 -> Modes = 2 and 3
Multimodal	More than two modes	1,1,2,2,3,3 -> Modes = 1, 2, 3

Type	Meaning	Example
No Mode	All values appear the same number of times	1, 2, 3, 4, 5 -> No Mode

Detailed Example 1 - Ungrouped Data

Data: 5, 8, 9, 8, 6, 8, 7, 5, 8, 10, 3, 8

Frequency count: 3(x1), 5(x2), 6(x1), 7(x1), 8(x5), 9(x1), 10(x1)

Mode = 8 (appears 5 times, more than any other value)

Detailed Example 2 - Business Application (Retail)

A shoe store recorded the following shoe sizes sold in one day: 5, 6, 7, 8, 8, 9, 8, 7, 8, 10, 6, 8, 9, 7, 8, 6, 7, 8

Shoe Size	Pairs Sold
Size 5	1 pair
Size 6	3 pairs
Size 7	4 pairs
Size 8	7 pairs (MOST FREQUENT)
Size 9	2 pairs
Size 10	1 pair

Mode = Size 8 (sold 7 times - highest frequency)

Business Decision: The store manager should stock Size 8 in the largest quantity to meet demand. Mode is the most useful measure here as it directly tells the most popular product.

Merits and Demerits of Mode

Type	Points
MERITS	Easy to understand; Not affected by extreme values; Can be used for qualitative data; Represents the most common value; Useful in business for popular products.
DEMERITS	May not be uniquely defined (bimodal or multimodal); Does not use all values; Affected by grouping of data; Not suitable for further algebraic calculations.

Q1 Define: Independent Events, Mutually Exclusive Events, Equally Likely Events
6

1. Independent Events

Two events A and B are said to be **Independent Events** if the occurrence or non-occurrence of one event **does not affect the probability** of the other event occurring. Each event is entirely unrelated to the other.

$$\text{Condition: } P(A \text{ and } B) = P(A) \times P(B)$$

$$\text{Also: } P(A | B) = P(A) \text{ and } P(B | A) = P(B)$$

Example 1 (Classic): Tossing a coin and rolling a die simultaneously.

$$P(\text{Head}) = 1/2; P(\text{Getting 4 on die}) = 1/6$$

$$P(\text{Head AND 4}) = 1/2 \times 1/6 = 1/12 \rightarrow \text{Events are independent.}$$

Example 2 (Business): Selecting a defective item from Batch A and selecting a defective item from Batch B.

If batches are from different sources, defects are independent of each other.

Example 3: Rain today and the result of a cricket match in another city are independent events.

2. Mutually Exclusive Events

Two or more events are said to be **Mutually Exclusive** (or Disjoint) if they **cannot occur simultaneously**. The occurrence of one event completely prevents the occurrence of the other. There is no common outcome between them.

$$\text{Condition: } P(A \text{ and } B) = P(A \text{ intersection } B) = 0$$

$$\text{Addition Rule: } P(A \text{ or } B) = P(A) + P(B) \text{ [since } P(A \text{ and } B) = 0]$$

Example 1 (Coin): Getting a Head (H) and getting a Tail (T) in a SINGLE toss.

$$P(H \text{ and } T) = 0 \rightarrow \text{Cannot get both Head and Tail in one toss.}$$

$$P(H \text{ or } T) = P(H) + P(T) = 1/2 + 1/2 = 1$$

Example 2 (Die): Getting an Even number and getting an Odd number in a single roll.

{2,4,6} and {1,3,5} share no common elements $\rightarrow$ Mutually Exclusive.

Example 3 (Business): A product being classified as 'Defective' or 'Non-defective'. It cannot be both at once.

3. Equally Likely Events

Events are said to be **Equally Likely** if each event has an **equal chance (probability) of occurring**. None of the outcomes is more likely than any other. This is the basis of classical (a priori) probability.

$$P(\text{each event}) = 1 / n \text{ where } n = \text{total number of equally likely outcomes}$$

Example 1 (Coin): Tossing a FAIR coin - $P(\text{Head}) = P(\text{Tail}) = 1/2 \rightarrow$ Both outcomes equally likely.

Example 2 (Die): Rolling a FAIR die - $P(1)=P(2)=P(3)=P(4)=P(5)=P(6) = 1/6 \rightarrow$ All 6 faces equally likely.

Example 3 (Cards): Drawing any single card from a well-shuffled deck of 52 cards.

$P(\text{any specific card}) = 1/52 \rightarrow$ All 52 cards are equally likely to be drawn.

Example 4 (Business): If a lucky draw has 100 equally numbered tickets, each person has $P = 1/100$.

Summary Comparison

Type	Key Condition	Example
Independent Events	Occurrence of one does not affect the other	Tossing coin + rolling die
Mutually Exclusive Events	Cannot occur at the same time; no overlap	Head or Tail in one toss
Equally Likely Events	Each has the same probability of occurring	All faces of a fair die

Q1 7 Define Hypothesis

Definition

A **Hypothesis** is a **precise, testable, and tentative statement or assumption about a population parameter or about the relationship between two or more variables**, which can be accepted or rejected based on statistical evidence gathered from sample data. The word 'Hypothesis' comes from Greek meaning 'under supposition'.

Definitions by Scholars

Scholar	Definition
Kerlinger (1986)	A hypothesis is a conjectural statement of the relation between two or more variables.
Goode and Hatt	A hypothesis is a proposition which can be put to a test to determine its validity.
Black and Champion	A hypothesis is a tentative statement about something, the validity of which is usually unknown.

Nature and Characteristics of a Good Hypothesis

- (i) Clear and Precise: The hypothesis must be stated in clear, specific, and unambiguous language.
- (ii) Testable and Verifiable: It must be possible to collect data and test the hypothesis statistically.
- (iii) Related to Theory: A good hypothesis is consistent with existing knowledge and established theory.
- (iv) Limited in Scope: It should be narrow enough to test, not too broad or vague.
- (v) Stated in Simple Terms: It should use simple, operational language that can be measured.
- (vi) Falsifiable: It must be possible for the hypothesis to be proven wrong (Popper's criterion).
- (vii) Relevant: It should be directly related to the research problem being investigated.

The Two Core Hypotheses Used in Testing

Hypothesis	Meaning	Example
Null Hypothesis (H ₀)	Statement of NO effect, NO difference, or NO relationship. It is always assumed true until evidence proves otherwise.	H ₀ : There is no difference in average sales before and after the campaign. ($m_1 = m_2$)
Alternative Hypothesis (H ₁)	Statement that contradicts H ₀ . Represents what the researcher expects to find or prove. H ₁ is accepted when H ₀ is rejected.	H ₁ : Average sales INCREASED after the campaign. ($m_1 < m_2$)

Q1
8

Explain Steps in Formulation of Hypothesis

Definition

Formulation of hypothesis is the **systematic process of constructing a clear, testable, and meaningful statement about the expected relationship or difference in the population**. It is one of the most critical steps in the research process, as a poorly formulated hypothesis can lead to inconclusive or misleading results.

Steps in Formulation of Hypothesis

Step No.	Step Name	Action / Description	Example
Step 1	Identify and Define the Research Problem	The researcher must clearly identify the problem that needs investigation. A well-defined problem is the foundation of a good hypothesis.	Problem: Does a new fertiliser increase wheat yield compared to the old one?
Step 2	Review of Existing Literature	Study previous research, journals, books, and theories related to the topic. This helps avoid duplication and provides a strong theoretical base.	Read agricultural journals, past experiments on fertilisers, and crop yield studies.

Step No.	Step Name	Action / Description	Example
Step 3	Identify Variables	Determine the Dependent Variable (outcome you measure) and Independent Variable (cause or input). Extraneous variables should also be identified and controlled.	IV: Type of fertiliser used. DV: Wheat yield (kg per acre)
Step 4	State the Null Hypothesis (H0)	Formulate H0 by assuming there is NO significant difference or effect. H0 is always an equality statement and represents the status quo.	H0: $m_1 = m_2$ (New fertiliser yield = Old fertiliser yield; no difference)
Step 5	State the Alternative Hypothesis (H1)	Formulate H1 as the statement you want to prove; the opposite of H0. Decide whether it is one-tailed (directional) or two-tailed (non-directional).	H1: $m_1 > m_2$ (New fertiliser gives higher yield; one-tailed, directional)
Step 6	Specify Level of Significance (α)	Set the acceptable risk of rejecting a true H0 (Type I error). Common levels: 5% ($\alpha=0.05$) or 1% ($\alpha=0.01$). Lower α = more stringent test.	$\alpha = 0.05$ means we accept 5% chance of wrongly concluding the fertiliser is better.
Step 7	Express Hypothesis in Statistical Terms	Translate the hypothesis into mathematical symbols using population parameters. This makes the hypothesis precise and testable using formulas.	H0: $m_1 - m_2 = 0$ vs. H1: $m_1 - m_2 > 0$
Step 8	Ensure Testability and Feasibility	Verify that the hypothesis can be tested with data that can realistically be collected within available time, budget, and ethical constraints.	Can we run a field experiment on 50 farmers for one season? Yes -> Feasible.

Complete Worked Example - Sales Training

Research Problem: Does a new 2-week sales training programme improve employee sales performance?

Step 1: Problem = Effect of training on sales performance.

Step 2: Literature shows similar programmes improved performance by 10-20% in other companies.

Step 3: IV = Training (Yes/No). DV = Average sales per week (in Rs.).

Step 4: H0: $m(\text{after}) = m(\text{before}) = \text{Rs.}50,000$ [Training has NO effect]

Step 5: H1: $m(\text{after}) > \text{Rs.}50,000$ [Training INCREASES sales; one-tailed]

Step 6: Set $\alpha = 0.05$ (5% significance level)

Step 7: $H_0: m \geq 50000$ vs $H_1: m < 50000$

Step 8: Test using paired t-test on before and after sales data of 30 employees.

Key Points to Remember

- (i) H_0 always uses equality signs: $=$, $\geq$, $\leq$
 - (ii) H_1 always uses inequality signs: $\neq$, $>$, $<$
 - (iii) The hypothesis should be formulated BEFORE collecting data.
 - (iv) A rejected H_0 does not prove H_1 ; it only provides evidence in favour of H_1 .
 - (v) Statistical conclusions are probabilistic, not absolute certainties.
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